Project Design Phase-II
Technology Stack (Architecture & Stack)

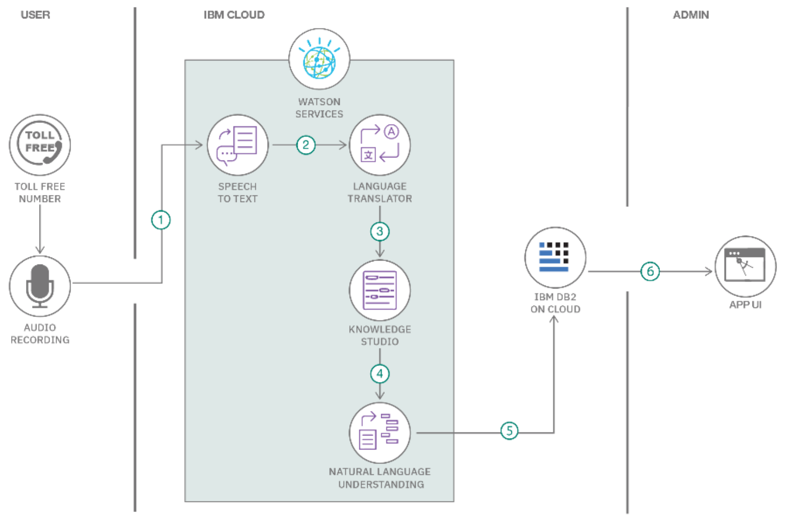
Date	31 January 2026
Team ID	LTVIP2026TMIDS35350
Project Name	Empowering India
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1: Components & Technologies:

S.No	Component	Description	Technology Used
1.	User Interface	Web-based analytics dashboard where users interact with budget visualizations, filters, and story navigation.	HTML5, CSS3, JavaScript, Flask Templates
2.	Application Logic-1	Backend logic to render webpages and integrate Tableau dashboards into Flask application.	Python (Flask Framework)
3.	Application Logic-2	Data visualization logic and analytical computations performed on Union Budget dataset.	Tableau Desktop
4.	Application Logic-3	Interactive storytelling, filtering, and dashboard navigation logic.	Tableau Story & Dashboard Features
5.	Database	Dataset containing Union Budget allocations, departments, schemes, and financial metrics.	CSV File (Structured Dataset)
6.	Cloud Database	Public hosting of visual analytics dashboards for global access.	Tableau Public Cloud
7.	File Storage	Local storage of dataset, Flask templates, and project files.	Local File System
8.	External API-1	Tableau Public Embed API for integrating dashboards into web application.	Tableau JavaScript API
9.	External API-2	Font and UI enhancement services for professional interface design.	Google Fonts API, Font Awesome CDN
10.	Machine Learning Model	(Not used in this project) — purely data analytics and visualization based.	Not Applicable
11.	Infrastructure (Server / Cloud)	Deployment of Flask web application locally or on cloud platform.	Local Server / Render / Railway / Cloud Deployment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The application uses open-source technologies for data processing, visualization, and web integration to ensure cost-effectiveness and flexibility.	Python, Flask, MySQL, Tableau Public
2.	Security Implementations	The system ensures secure access through role-based authentication and encrypted communication. Data validation and secure database connections are implemented to prevent unauthorized access.	HTTPS, Role-Based Access Control (RBAC), SHA-256 Password Hashing, OWASP Best Practices
3.	Scalable Architecture	The application follows a 3-tier architecture (Presentation Layer, Application Layer, Database Layer). It supports future expansion such as adding more years of budget data or additional dashboards.	Flask (Backend), Tableau (Visualization Layer), MySQL (Database)
4.	Availability	The system ensures high availability by hosting the dashboard online and maintaining reliable database connectivity. Backup mechanisms ens-ure data safety	Tableau Public Hosting, Cloud Deployment, Database Backup Mechanism
5.	Performance	The system is optimized for fast query execution and dashboard loading. SQL indexing and efficient filtering improve response time even with large datasets.	MySQL Indexing, Optimized SQL Queries, Tableau Data Extracts

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d